The sharp A_p bound for the Hardy–Littlewood maximal operator

An exposition of Lerner's proof of Buckley's theorem

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Abstract

B. Muckenhoupt [6] showed that the Hardy–Littlewood maximal operator is bounded on $L^p_w(\mathbb{R}^d)$, $1 < p < \infty$, if and only if $w \in A_p$. S. Buckley [1] proved that its operator norm is at most a constant times $[w]_{A_p}^{1/(p-1)}$, and that this exponent is best possible. This report presents A. K. Lerner's later proof [5] of this bound, together with Buckley's example showing that the exponent cannot be improved.

1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$ that is not almost everywhere zero*[1]*. For a measurable function $v \geq 0$ on $\mathbb{R}^d$ and a measurable set $E \subseteq \mathbb{R}^d$, $v(E) := \int_E v(y)\,dy \in [0,\infty]$. The map $E \mapsto v(E)$ is a measure on the Lebesgue measurable sets.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \leq p < \infty$. The weighted L^p space is*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(y)|^p\, w(y)\,dy < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(y)|^p\, w(y)\,dy\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

The central object of this report is the Hardy–Littlewood maximal operator M. The proofs also use its analogue with respect to the measure $w(y)\,dy$.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

[1] If $w = 0$ almost everywhere, then $\int_{\mathbb{R}^d} |f|^p w\,dy = 0$ for every measurable f, and the boundedness of M holds trivially. This is the same convention as used in [1, Section 1].

where the supremum is taken over all closed cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. All cubes in this report are of this kind. If the supremum is taken only over the cubes $Q(x,r) := \{y : |y-x|_\infty \le r/2\}$, $r > 0$, where $|z|_\infty := \max_{1\le k\le d} |z_k|$, the operator is denoted M_w^c. If $w \equiv 1$, the subscript is dropped, giving M and M^c. The Euclidean norm of $z \in \mathbb{R}^d$ is denoted $|z|$, and $B(x,r) := \{y : |y-x| < r\}$.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) for $1 < p \le \infty$ [4, Section 2.1], [3, Theorem 1.1], and Section 2.2 adapts the standard proof to weighted maximal operators. Weighted versions of these bounds are used to prove weighted mean convergence and summability results for orthogonal series, and they are needed to characterise the weights for which the conjugate function and the Hilbert transform are bounded [6, Section 1]. For $1 < p < \infty$ and a weight w, M is bounded on $L^p_w(\mathbb{R}^d)$ if there is a constant C such that for all $f \in L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w\, dy\right)^{1/p} \le C \left(\int_{\mathbb{R}^d} |f|^p \, w\, dy\right)^{1/p} = C\, \|f\|_{L^p_w}. \tag{1}$$

If (1) holds, then each $f \in L^p_w(\mathbb{R}^d)$ and each cube Q give a condition on w, and Muckenhoupt showed that a suitable choice of f for each cube gives the following necessary condition. It is not needed for the proof of the main result, but it motivates the definition of A_p below, so its proof is included in Appendix A.

Proposition 1 ([6, Theorem 9], [3, proof of Theorem 1.7])**.** *Let $1 < p < \infty$ and let w be a weight, with $w^{-\frac{1}{p-1}} := \infty$ on $\{w = 0\}$. If (1) holds with constant C, then every cube Q satisfies $\int_Q w^{-\frac{1}{p-1}}\, dy < \infty$ and*

$$\left(\frac{1}{|Q|}\int_Q w\, dy\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{p-1} \le C^p. \tag{2}$$

In particular, $w > 0$ almost everywhere.

Definition 4 (Muckenhoupt A_p weights [6, (7.10)])**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(y)\, dy\right)\left(\frac{1}{|Q|}\int_Q w(y)^{-\frac{1}{p-1}}\, dy\right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$, with the convention $0 \cdot \infty := 0$ [6, Section 1]. The weight w belongs to A_p if $[w]_{A_p} < \infty$.

Definition 5 (Dual weight [1, Section 1])**.** *Let w be a weight and let $1 < p < \infty$. The* dual weight *of w is $\sigma := w^{-\frac{1}{p-1}}$ (not necessarily a weight in the sense of Definition 1). With this notation (3) takes the more compact form*

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|}\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight. If (1) holds with constant C, then $w \in A_p$ with*

$$[w]_{A_p} \le C^p. \tag{4}$$

Muckenhoupt proved that the condition $w \in A_p$ is also sufficient.

Theorem 1 (Muckenhoupt [6, Theorem 9]). *Let $1 < p < \infty$ and let w be a weight. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem 1 shows that the operator norm of M on $L^p_w(\mathbb{R}^d)$, that is, the smallest constant C in (1), is finite precisely when $w \in A_p$. By Corollary 1, this norm is bounded from below by $[w]^{1/p}_{A_p}$. The natural question is therefore whether it is also bounded from above by a power of $[w]_{A_p}$, and which power is the smallest possible.

The answer matters beyond M. The dependence of operator norms on $[w]_{A_p}$ later became a central question for singular integrals such as the Hilbert transform, known as the A_2 conjecture, and M was the first operator for which it was determined [7].

Theorem 2 (Buckley [1, Theorem 2.5], Lerner [5, Theorem C]). *Let $1 < p < \infty$.*

(i) There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,

$$\|Mf\|_{L^p_w} \le c_{p,d}\,[w]^{\frac{1}{p-1}}_{A_p}\,\|f\|_{L^p_w}.$$

(ii) If $\alpha < \frac{1}{p-1}$, then there is no constant c such that $\|Mf\|_{L^p_w} \le c\,[w]^{\alpha}_{A_p}\,\|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.

Buckley's proof of Theorem 2(i) [1, proof of Theorem 2.5], like Muckenhoupt's proof of Theorem 1 [6, Lemma 5 and proof of Theorem 9], relies on the deep property that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$ [3, Theorem 4.5], together with interpolation. Lerner [5, Theorem C] found a short and elementary proof of Theorem 2(i). Theorem 2 is Theorem C in [5], whose proof there covers only part (i). This report presents his argument in detail and completes it with a proof of part (ii), based on Buckley's example in $\mathbb{R}$ [1, proof of Theorem 2.5] and extended here to $\mathbb{R}^d$.

Remark 1 (An alternative proof of Theorem 1). *If M is bounded on $L^p_w(\mathbb{R}^d)$, then $w \in A_p$ by Corollary 1, which follows from Proposition 1. Conversely, if $w \in A_p$, then M is bounded on $L^p_w(\mathbb{R}^d)$ by Theorem 2(i). Combined, these results imply Theorem 1.*

2 Preliminaries

2.1 Basic results on cubes, weights and maximal operators

Lemma 1. *Let Q be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z - y|_\infty \le \ell(Q)$, since both z and y lie in Q. Hence $Q \subseteq Q(y, 2\ell(Q))$. If c is the centre of Q, then $|y - c|_\infty \le \frac{1}{2}\ell(Q)$, so every $z \in Q(y, 2\ell(Q))$ satisfies $|z - c|_\infty \le |z - y|_\infty + |y - c|_\infty \le \frac{3}{2}\ell(Q)$, that is, $z \in 3Q$. □

Lemma 2 ([4, Section 2.1.1]). *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \le 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and $|Q(x, 2\ell(Q))| = 2^d|Q|$ gives

$$\begin{aligned} Mf(x) &= \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\, dy \leq \sup_{Q \ni x} \frac{2^d}{|Q(x, 2\ell(Q))|} \int_{Q(x,2\ell(Q))} |f(y)|\, dy \\ &\leq 2^d \sup_{r>0} \frac{1}{|Q(x,r)|} \int_{Q(x,r)} |f(y)|\, dy = 2^d M^c f(x). \end{aligned}$$

□

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$. Then $w > 0$ almost everywhere and σ, as in Definition 5, is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q, and $L^p_w(\mathbb{R}^d) \subseteq L^1_{\mathrm{loc}}(\mathbb{R}^d)$.*

Proof. Let Q be a cube with $w(Q) > 0$. By the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty, \tag{5}$$

so $\sigma(Q) < \infty$ on any cube Q where $w(Q) > 0$.

Assume, for contradiction, that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, $w > 0$ on a set F with $|F| > 0$, so some cube Q satisfies $|Q \cap E| > 0$ and $|Q \cap F| > 0$. Then $w(Q) \geq w(Q \cap F) > 0$, so $\sigma(Q) < \infty$ by (5). But $\sigma = \infty$ on $Q \cap E$ by Definition 1, so $\sigma(Q) = \infty$, a contradiction. Therefore $w > 0$ almost everywhere. In particular $w(Q) > 0$, and hence $\sigma(Q) < \infty$ by (5), for every cube Q.

Every cube Q has $w(Q) < \infty$, so $w < \infty$ almost everywhere. Hence $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. As every compact set lies in a cube, σ is locally integrable and hence a weight. Finally, for $f \in L^p_w(\mathbb{R}^d)$ and every cube Q, applying Hölder's inequality gives

$$\int_Q |f|\, dy = \int_Q |f|\, w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\, dy \leq \Big(\int_Q |f|^p\, w\, dy \Big)^{\frac{1}{p}} \Big(\int_Q w^{-\frac{1}{p-1}}\, dy \Big)^{\frac{p-1}{p}} \leq \|f\|_{L^p_w}\, \sigma(Q)^{\frac{p-1}{p}},$$

which is finite. □

2.2 Weight-independent bounds for M^c_v

For M, the Vitali covering lemma gives the weak type $(1,1)$ estimate, and Marcinkiewicz interpolation with the bound on L^∞ gives the strong type (p,p) estimate [4, Sections 1.3.1 and 2.1.1]. The Vitali covering lemma enlarges the selected cubes Q to $3Q$, so for the measure $v(y)\, dy$ the constant depends on $v(3Q)/v(Q)$, which for $v \in A_p$ is only bounded in terms of $[v]_{A_p}$ [3, Remark 1.11]. The Besicovitch covering lemma, by contrast, does not enlarge the cubes, but it requires them to be centred. Used in place of the Vitali covering lemma, it gives the analogous estimates for M^c_v by the same arguments, with constants depending only on d and q (Lemmas 5 and 6).

Lemma 4 (Besicovitch covering lemma [2, Theorem 1.1])**.** *Let $E \subset \mathbb{R}^d$ be a bounded set, and suppose that for each $x \in E$ a cube $Q(x, r_x)$ centred at x is given. Then there are points $x_1, x_2, \ldots \in E$ such that the cubes $Q(x_j, r_{x_j})$ cover E and every point of $\mathbb{R}^d$ lies in at most N_d of them, where N_d depends only on d.*

Proof. Assume that E is nonempty and let $D := \sup_{x,y\in E} |x-y|_\infty < \infty$. If $r_x \geq 2D$ for some $x \in E$, then $E \subseteq Q(x, r_x)$ and this single cube suffices. Hence assume that $r_x < 2D$ for every $x \in E$.

Let $s_0 := \sup_{x\in E} r_x$ and choose $x_1 \in E$ with $r_{x_1} > s_0/2$. Given $x_1, \dots, x_m$, let

$$s_m := \sup\Big\{ r_x : x \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j}) \Big\}$$

and choose $x_{m+1} \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})$ with $r_{x_{m+1}} > s_m/2$. The selection stops if this set is empty. If $j < i$, then x_i lies outside $Q(x_j, r_{x_j})$, and x_i was a possible choice when x_j was chosen. Hence

$$|x_i - x_j|_\infty > \frac{r_{x_j}}{2} \qquad \text{and} \qquad r_{x_i} \leq s_{j-1} < 2r_{x_j}. \tag{6}$$

If the selection stops, the cubes $Q(x_j, r_{x_j})$ cover E by construction. Assume that it does not stop, and suppose, for contradiction, that some $x \in E$ is not covered. Then x is a possible choice at every step, so $r_x \leq s_{j-1} < 2r_{x_j}$ for every j. If $j < i$, then by (6)

$$\frac{r_{x_i}}{8} + \frac{r_{x_j}}{8} < \frac{3r_{x_j}}{8} < |x_i - x_j|_\infty,$$

so the cubes $Q(x_j, r_{x_j}/4)$ are pairwise disjoint. Since $x_j \in E$ and $r_{x_j} < 2D$, they are all contained in $Q(x_1, 4D)$. Comparing volumes, for every N,

$$N\left(\frac{r_x}{8}\right)^d < \sum_{j=1}^{N} \left(\frac{r_{x_j}}{4}\right)^d \leq (4D)^d,$$

which is false for large N. Hence every $x \in E$ is covered.

Fix $z \in \mathbb{R}^d$. For $\varepsilon \in \{-1, 1\}^d$ let

$$O_\varepsilon := \{y \in \mathbb{R}^d : \varepsilon_k(y_k - z_k) \geq 0 \text{ for } k = 1, \dots, d\}.$$

Every $y \in \mathbb{R}^d$ lies in O_ε with $\varepsilon_k := 1$ if $y_k \geq z_k$ and $\varepsilon_k := -1$ otherwise, so these 2^d orthants cover $\mathbb{R}^d$. If $y, y' \in O_\varepsilon$, then $y_k - z_k$ and $y'_k - z_k$ have the same sign for every k, and hence

$$|y - y'|_\infty \leq \max\{|y - z|_\infty,\ |y' - z|_\infty\}. \tag{7}$$

For each ε let

$$J_\varepsilon := \{x_i : x_i \in O_\varepsilon \text{ and } z \in Q(x_i, r_{x_i})\}.$$

Fix ε_0 with J_{ε_0} nonempty and let $j := \min\{i : x_i \in J_{\varepsilon_0}\}$, so that x_j is the point of J_{ε_0} that was chosen first. Let $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$, so that $i > j$. Since z lies in both $Q(x_i, r_{x_i})$ and $Q(x_j, r_{x_j})$, (6) and (7) give

$$\frac{r_{x_j}}{2} < |x_i - x_j|_\infty \leq \max\{|x_i - z|_\infty,\ |x_j - z|_\infty\} \leq \max\Big\{\frac{r_{x_i}}{2},\ \frac{r_{x_j}}{2}\Big\}.$$

Hence $r_{x_i} > r_{x_j}$, and by (6) also $|x_i - z|_\infty \leq r_{x_i}/2 < r_{x_j}$.

For each $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$ consider the cube $Q(x_i, r_{x_j}/2)$. If $x_i, x_m \in J_{\varepsilon_0} \setminus \{x_j\}$ with $m < i$, then $r_{x_m} > r_{x_j}$, so (6) gives $|x_i - x_m|_\infty > r_{x_m}/2 > r_{x_j}/2$. Thus the centres of

these cubes are more than their side length apart, and the cubes are pairwise disjoint. Moreover, for $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$ every $y \in Q(x_i, r_{x_j}/2)$ satisfies

$$|y - z|_\infty \le |y - x_i|_\infty + |x_i - z|_\infty < \frac{r_{x_j}}{4} + r_{x_j} = \frac{5r_{x_j}}{4},$$

so all of them are contained in $Q(z, 5r_{x_j}/2)$. If n of these cubes are chosen, then, since they are disjoint, their total volume is at most the volume of $Q(z, 5r_{x_j}/2)$, that is,

$$n\left(\frac{r_{x_j}}{2}\right)^d \le \left(\frac{5r_{x_j}}{2}\right)^d = 5^d\left(\frac{r_{x_j}}{2}\right)^d.$$

Hence $n \le 5^d$, so J_{ε_0} has at most $1 + 5^d$ points. Since ε_0 was an arbitrary choice with J_{ε_0} nonempty, every J_ε has at most $1 + 5^d$ points. Every cube $Q(x_i, r_{x_i})$ containing z has its centre in some O_ε, and hence in J_ε. Summing over the 2^d orthants, z lies in at most $2^d(1 + 5^d)$ of the cubes $Q(x_i, r_{x_i})$, and the lemma holds with $N_d := 2^d(1 + 5^d)$. □

Lemma 5 ([3, Theorem 1.10])**.** *Let v be a weight with $v(Q) > 0$ for every cube Q, and let f be a measurable function with $|f|\, v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $M^c_v f$ is measurable, and for every $t > 0$,*

$$v\big(\{x \in \mathbb{R}^d : M^c_v f(x) > t\}\big) \le \frac{N_d}{t}\,\|f\|_{L^1_v},$$

where N_d is the constant in Lemma 4.

Proof. Let $r > 0$ and $x \in \mathbb{R}^d$, and let (x_n) be a sequence converging to x. If $|y - x|_\infty < r/2$, then $|y - x_n|_\infty < r/2$ for all large n. If $|y - x|_\infty > r/2$, then $|y - x_n|_\infty > r/2$ for all large n. Hence $\chi_{Q(x_n,r)}(y) \to \chi_{Q(x,r)}(y)$ for every y outside the boundary of $Q(x, r)$, which is a set of measure zero. Moreover, $R := \sup_n |x_n - x|_\infty$ is finite, and $Q(x_n, r) \subseteq Q(x, r + 2R)$ for every n, so $\chi_{Q(x_n,r)}\,|f|\, v \le \chi_{Q(x,r+2R)}\,|f|\, v$, which is integrable since $|f|\, v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. By dominated convergence,

$$\lim_{n\to\infty}\int_{Q(x_n,r)} |f|\, v\, dy = \lim_{n\to\infty}\int_{\mathbb{R}^d} \chi_{Q(x_n,r)}\,|f|\, v\, dy = \int_{\mathbb{R}^d} \chi_{Q(x,r)}\,|f|\, v\, dy = \int_{Q(x,r)} |f|\, v\, dy.$$

Thus $x \mapsto \int_{Q(x,r)} |f|\, v\, dy$ is sequentially continuous, and hence continuous. Taking $f \equiv 1$ shows that $x \mapsto v(Q(x, r))$ is continuous as well. Since $v(Q(x, r)) > 0$, the average

$$x \mapsto \frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f|\, v\, dy$$

is continuous for every $r > 0$. For $t > 0$ the set $\{M^c_v f > t\}$ is the union over $r > 0$ of the sets where this average exceeds t. These are open, so $\{M^c_v f > t\}$ is open. Since $t > 0$ was arbitrary and $M^c_v f \ge 0$, the function $M^c_v f$ is measurable.

Now fix $t > 0$. For $k = 1, 2, \ldots$ let $E_k := \{x : |x|_\infty < k \text{ and } M^c_v f(x) > t\}$. Let $x \in E_k$. Since $M^c_v f(x) > t$, the definition of M^c_v gives some $r > 0$ with $\frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f|\, v\, dy > t$. Choose such an r and call it r_x. Since E_k is bounded, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover E_k and $\sum_j \chi_{Q(x_j,r_{x_j})} \le N_d$. Hence

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{t}\sum_j \int_{Q(x_j,r_{x_j})} |f|\, v\, dy \\ &= \frac{1}{t}\int_{\mathbb{R}^d} \Big(\sum_j \chi_{Q(x_j,r_{x_j})}\Big)|f|\, v\, dy \le \frac{N_d}{t}\,\|f\|_{L^1_v}, \end{aligned}$$

where the second inequality holds by the choice of r_{x_j}, and the equality follows from monotone convergence. Since $E_1 \subseteq E_2 \subseteq \cdots$ and v is a measure, the bound for $v(E_k)$ gives

$$v\big(\{M_v^c f > t\}\big) = v\Big(\bigcup_k E_k\Big) = \lim_{k\to\infty} v(E_k) \le \frac{N_d}{t}\,\|f\|_{L^1_v}.$$

□

Lemma 6 ([3, Theorem 1.10])**.** *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M_v^c f\|_{L^q_v} \le C_{q,d}\,\|f\|_{L^q_v}$$

for all $f \in L^q_v(\mathbb{R}^d)$.

Given Lemma 5, the proof of Lemma 6 is the standard interpolation argument for M with dy replaced by $v(y)\,dy$ [4, Section 2.1.1], and it is therefore given in Appendix B.

3 Proof of Theorem 2

Lemma 7 (Lerner [5, proof of Theorem C])**.** *Let $1 < p < \infty$, let $w \in A_p$, and let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then, for every $x \in \mathbb{R}^d$,*

$$M^c f(x) \le 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}} \Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{8}$$

Proof. By Lemmas 3 and 5, σ is a weight, M_w^c and M_σ^c are defined, and $M_\sigma^c(|f|\sigma^{-1})$ is measurable. Let $x \in \mathbb{R}^d$ and $r > 0$. By Lemma 1, $Q(x,r) \subseteq Q(x,3r)$, and $|Q(x,3r)| = 3^d\,|Q(x,r)|$. Together with Definition 5 this implies

$$\frac{w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}}{|Q(x,r)|^p} \le 3^{dp}\,\frac{w(Q(x,3r))}{|Q(x,3r)|}\left(\frac{\sigma(Q(x,3r))}{|Q(x,3r)|}\right)^{p-1} \le 3^{dp}\,[w]_{A_p}. \tag{9}$$

Every $z \in Q(x,r)$ satisfies $Q(x,r) \subseteq Q(z,2r) \subseteq Q(x,3r)$ by Lemma 1. Together with Definition 3 this implies

$$\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)} |f|\,\sigma^{-1}\sigma\,dy \le \frac{1}{\sigma(Q(z,2r))}\int_{Q(z,2r)} |f|\,\sigma^{-1}\sigma\,dy \le M_\sigma^c\big(|f|\sigma^{-1}\big)(z). \tag{10}$$

Multiplying by 1 then gives

$$
\begin{aligned}
&\frac{1}{|Q(x,r)|}\int_{Q(x,r)}|f|\,dy\\
&\quad=\Bigg(\underbrace{\frac{w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}}{|Q(x,r)|^p}}_{\le 3^{dp}[w]_{A_p}\text{ by (9)}}\frac{|Q(x,r)|^p}{w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}}\Bigg)^{\frac{1}{p-1}}\frac{1}{|Q(x,r)|}\int_{Q(x,r)}|f|\,dy\\
&\quad\le 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\left(\frac{|Q(x,r)|}{w(Q(x,r))}\left(\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)}|f|\,\sigma^{-1}\sigma\,dy\right)^{p-1}\right)^{\frac{1}{p-1}}\\
&\quad=3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\left(\frac{1}{w(Q(x,r))}\int_{Q(x,r)}\Bigg(\underbrace{\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)}|f|\,\sigma^{-1}\sigma\,dy}_{\le M_\sigma^c(|f|\sigma^{-1})(z)\text{ for all }z\in Q(x,r)\text{ by (10)}}\Bigg)^{p-1}dz\right)^{\frac{1}{p-1}}\\
&\quad\le 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\left(\frac{1}{w(Q(x,r))}\int_{Q(x,r)}\Big(M_\sigma^c\big(|f|\sigma^{-1}\big)(z)\Big)^{p-1}w(z)^{-1}\,w(z)\,dz\right)^{\frac{1}{p-1}}\\
&\quad\le 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}.
\end{aligned}
$$

Taking the supremum over $r>0$ gives (8). □

Proof of Theorem 2(i). Let $1<p<\infty$, let $w\in A_p$ and let $f\in L^p_w(\mathbb{R}^d)$. Lemma 3 gives $f\in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and $w(Q)>0$ and $\sigma(Q)>0$ for every cube Q. Lemma 6 therefore shows that M_w^c and M_σ^c are of strong type (q,q) on $L^q_w(\mathbb{R}^d)$ and $L^q_\sigma(\mathbb{R}^d)$, respectively, for every $1<q<\infty$. The sets $\{Mf>t\}$, $t>0$, are open [4, Section 2.1.1], so Mf is measurable. Lemma 5 with $v\equiv 1$ shows that M^cf is measurable as well.

The identities $w^{1-\frac{p}{p-1}}=\sigma$, $\sigma^{1-p}=w$, and Lemma 6 with $v=\sigma$ and $q=p$, give

$$
\Big\|\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big\|_{L^{p/(p-1)}_w}^{\frac{1}{p-1}}=\big\|M_\sigma^c\big(|f|\sigma^{-1}\big)\big\|_{L^p_\sigma}\le C_{p,d}\,\big\||f|\sigma^{-1}\big\|_{L^p_\sigma}=C_{p,d}\,\|f\|_{L^p_w}.\tag{11}
$$

In particular $\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\in L^{p/(p-1)}_w(\mathbb{R}^d)$. Lemma 2 gives the first inequality below, and bounding the marked factors as indicated gives

$$
\begin{aligned}
\|Mf\|_{L^p_w}&\le 2^d\Big\|\underbrace{M^cf}_{\text{bounded by Lemma 7}}\Big\|_{L^p_w}\\
&\le 2^d\Big\|3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big)\Big)^{\frac{1}{p-1}}\Big\|_{L^p_w}\\
&=2^d\,3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\underbrace{\Big\|M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big)\Big\|_{L^{p/(p-1)}_w}^{\frac{1}{p-1}}}_{\text{bounded by Lemma 6 with }v=w,\ q=\frac{p}{p-1}}\\
&\le 2^d\,3^{\frac{dp}{p-1}}\,C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\underbrace{\Big\|\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big\|_{L^{p/(p-1)}_w}^{\frac{1}{p-1}}}_{\text{bounded by (11)}}\\
&\le 2^d\,3^{\frac{dp}{p-1}}\,C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\,C_{p,d}\,\|f\|_{L^p_w}.
\end{aligned}
$$

The inequality in Theorem 2(i) therefore holds with $c_{p,d}:=2^d\,3^{\frac{dp}{p-1}}\,C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\,C_{p,d}$. □

Proof of Theorem 2(ii). Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant c such that

$$\|Mf\|_{L^p_w} \le c\,[w]_{A_p}^{\alpha}\,\|f\|_{L^p_w} \tag{12}$$

for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$. Let $w \in A_p$. Then $w > 0$ almost everywhere by Lemma 3, and for every cube Q, Hölder's inequality with exponents p and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|}\int_Q w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\,dy \le \left(\frac{1}{|Q|}\int_Q w\,dy\right)^{\frac{1}{p}} \left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Hence (12) remains true if α is replaced by $\max\{\alpha, 0\}$, so α may be assumed to satisfy $0 \le \alpha < \frac{1}{p-1}$. For $0 < \delta < 1$ let

$$w_\delta(x) := |x|^{(d-\delta)(p-1)} \qquad \text{and} \qquad f_\delta(x) := w_\delta(x)^{-\frac{1}{p-1}}\chi_{Q(0,2)}(x) = |x|^{\delta-d}\chi_{Q(0,2)}(x).$$

For every $r > 0$, polar coordinates give

$$\int_{B(0,r)} w_\delta(y)^{-\frac{1}{p-1}}\,dy = \int_{B(0,r)} |y|^{\delta-d}\,dy = |S^{d-1}|\int_0^r s^{\delta-d}\,s^{d-1}\,ds = |S^{d-1}|\,\frac{r^\delta}{\delta}, \tag{13}$$

where $|S^{d-1}|$ denotes the surface measure of the unit sphere $S^{d-1} \subseteq \mathbb{R}^d$.

Since $(d-\delta)(p-1) > 0$, the function w_δ is continuous on $\mathbb{R}^d$, hence locally integrable, and it is positive outside the origin, so it is a weight. Let $Q(x_0, \ell)$ be a cube. Every $y \in Q(x_0, \ell)$ satisfies $|y - x_0| \le \sqrt{d}\,|y - x_0|_\infty \le \frac{\sqrt{d}}{2}\ell$.

If $|x_0| \le \sqrt{d}\,\ell$, then $Q(x_0, \ell) \subseteq B(0, 2\sqrt{d}\,\ell)$. On this ball $w_\delta \le (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}$, and (13) gives

$$\begin{aligned}
&\left(\frac{1}{|Q(x_0,\ell)|}\int_{Q(x_0,\ell)} w_\delta\,dy\right)\left(\frac{1}{|Q(x_0,\ell)|}\int_{Q(x_0,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \\
&\quad\le (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}\left(\frac{1}{\ell^d}\int_{B(0,2\sqrt{d}\,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \\
&\quad= (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}\left(\frac{|S^{d-1}|\,(2\sqrt{d}\,\ell)^\delta}{\delta\,\ell^d}\right)^{p-1} = (2\sqrt{d})^{d(p-1)}\,|S^{d-1}|^{p-1}\,\delta^{1-p}.
\end{aligned}$$

If $|x_0| > \sqrt{d}\,\ell$, then $|y - x_0| < \frac{1}{2}|x_0|$, so $\frac{1}{2}|x_0| \le |y| \le \frac{3}{2}|x_0|$ for every $y \in Q(x_0, \ell)$. Bounding w_δ by $\left(\frac{3}{2}|x_0|\right)^{(d-\delta)(p-1)}$ and $|y|^{\delta-d}$ by $\left(\frac{1}{2}|x_0|\right)^{\delta-d}$ on $Q(x_0, \ell)$ gives

$$\left(\frac{1}{|Q(x_0,\ell)|}\int_{Q(x_0,\ell)} w_\delta\,dy\right)\left(\frac{1}{|Q(x_0,\ell)|}\int_{Q(x_0,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \le 3^{(d-\delta)(p-1)} \le 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes gives $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le K_1\,\delta^{1-p}, \qquad K_1 := \max\{(2\sqrt{d})^{d(p-1)}|S^{d-1}|^{p-1},\, 3^{d(p-1)}\}. \tag{14}$$

The norm of f_δ satisfies

$$\|f_\delta\|^p_{L^p_{w_\delta}} = \int_{Q(0,2)} \underbrace{|y|^{(\delta-d)p}}_{|f_\delta(y)|^p}\,\underbrace{|y|^{(d-\delta)(p-1)}}_{w_\delta(y)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \le \int_{B(0,2\sqrt{d})} |y|^{\delta-d}\,dy < \infty,$$

by $Q(0,2) \subseteq B(0,2\sqrt{d})$ and (13). Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, since the integrand is positive on $Q(0,2)$. Let $x \in Q(0,2)$ with $x \neq 0$, and let $r := |x|_\infty$, so that $0 < r \leq 1$ and $r \leq |x|$. The cube $Q(0,2r)$ contains x, is contained in $Q(0,2)$, where $f_\delta(y) = |y|^{\delta-d}$, and contains $B(0,r)$. The definition of M, together with bounding the marked factors, gives

$$\begin{aligned} Mf_\delta(x) &\geq \frac{1}{(2r)^d} \underbrace{\int_{Q(0,2r)} f_\delta \, dy}_{\geq \int_{B(0,r)} |y|^{\delta-d}\, dy} \geq \frac{1}{(2r)^d} \underbrace{\int_{B(0,r)} |y|^{\delta-d}\, dy}_{= |S^{d-1}|\, r^\delta/\delta \text{ by (13)}} \\ &= \frac{|S^{d-1}|}{2^d \delta} \underbrace{r^{\delta-d}}_{\geq |x|^{\delta-d},\ \text{as } \delta-d<0} \geq \frac{|S^{d-1}|}{2^d \delta} f_\delta(x). \end{aligned}$$

Since $f_\delta = 0$ outside $Q(0,2)$, this gives $Mf_\delta \geq \frac{|S^{d-1}|}{2^d\delta} f_\delta$ almost everywhere, and taking norms gives

$$\|Mf_\delta\|_{L^p_{w_\delta}} \geq K_2\, \delta^{-1} \|f_\delta\|_{L^p_{w_\delta}}, \qquad K_2 := 2^{-d} |S^{d-1}|. \tag{15}$$

Applying (12) to w_δ and f_δ, and dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$, gives together with (15), (14) and $\alpha \geq 0$

$$K_2\, \delta^{-1} \leq \frac{\|Mf_\delta\|_{L^p_{w_\delta}}}{\|f_\delta\|_{L^p_{w_\delta}}} \leq c\, [w_\delta]^\alpha_{A_p} \leq c\, K_1^\alpha\, \delta^{-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Hence

$$\delta^{\alpha(p-1)-1} \leq \frac{c\, K_1^\alpha}{K_2} \qquad \text{for every } 0 < \delta < 1.$$

The assumption $\alpha < \frac{1}{p-1}$ makes the exponent $\alpha(p-1)-1$ negative, so the left-hand side tends to ∞ as $\delta \to 0$, while the right-hand side is a constant. This is a contradiction. □

A Proof of Proposition 1

Proof. For every cube Q and $x \in Q$, $Mf(x) \geq \frac{1}{|Q|} \int_Q |f|\, dy$, so (1) gives

$$\Big(\frac{1}{|Q|} \int_Q |f|\, dy\Big)^p w(Q) \leq \int_Q (Mf)^p\, w\, dy \leq C^p \int_{\mathbb{R}^d} |f|^p\, w\, dy \tag{16}$$

for every cube Q and every $f \in L^p_w(\mathbb{R}^d)$. Now let Q be a cube. Since w is not almost everywhere zero, Q is contained in a cube Q_0 with $w(Q_0) > 0$. Then (16) with Q_0 in place of Q and $f = \chi_Q$ gives $(|Q|/|Q_0|)^p\, w(Q_0) \leq C^p\, w(Q)$, so $w(Q) > 0$.

For $n \geq 1$ let $f_n := \big(w + \frac{1}{n}\big)^{-\frac{1}{p-1}} \chi_Q$. Since $w \leq w + \frac{1}{n}$,

$$|f_n|^p\, w \leq \big(w + \tfrac{1}{n}\big)^{-\frac{p}{p-1}} \big(w + \tfrac{1}{n}\big) = f_n \leq n^{\frac{1}{p-1}},$$

so $f_n \in L^p_w(\mathbb{R}^d)$, and $w < \infty$ almost everywhere gives $\int_Q f_n\, dy > 0$. Then (16) with $f = f_n$, together with $|f_n|^p w \leq f_n$, gives

$$\Big(\frac{1}{|Q|} \int_Q f_n\, dy\Big)^p w(Q) \leq C^p \int_Q f_n\, dy.$$

The functions f_n increase to $w^{-\frac{1}{p-1}}\chi_Q$ as $n \to \infty$, so dividing by $\int_Q f_n\,dy$ and applying monotone convergence gives

$$C^p \geq \lim_{n\to\infty} \frac{w(Q)}{|Q|}\Big(\frac{1}{|Q|}\int_Q f_n\,dy\Big)^{p-1} = \frac{w(Q)}{|Q|}\Big(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\Big)^{p-1}.$$

Since $w(Q) > 0$, $\int_Q w^{-\frac{1}{p-1}}\,dy < \infty$. As Q was arbitrary, $w > 0$ almost everywhere. □

B Proof of Lemma 6

Proof. Let $f \in L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, so that $f = f_1 + f_2$ with $f_1 \in L^1_v(\mathbb{R}^d)$ and $f_2 \in L^\infty_v(\mathbb{R}^d)$. By the definition of $\|f_2\|_{L^\infty_v}$, the sets $\{|f_2| > \|f_2\|_{L^\infty_v} + 1/n\}$, $n = 1, 2, \ldots$, have v-measure zero, and hence so does their union $\{|f_2| > \|f_2\|_{L^\infty_v}\}$. This implies

$$\int_Q |f_2|\,v\,dy \leq \|f_2\|_{L^\infty_v}\,v(Q) \quad \text{for every cube } Q, \tag{17}$$

and therefore

$$\int_Q |f|\,v\,dy \leq \int_Q |f_1|\,v\,dy + \int_Q |f_2|\,v\,dy \leq \|f_1\|_{L^1_v} + \|f_2\|_{L^\infty_v}\,v(Q) < \infty.$$

Hence $|f|\,v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, so $M^c_v f$ is defined and, by Lemma 5, measurable.

Let $f, h \in L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, $x \in \mathbb{R}^d$ and $r > 0$. The pointwise inequality $|f + h| \leq |f| + |h|$ gives

$$\begin{aligned}\frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f+h|\,v\,dy &\leq \frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f|\,v\,dy + \frac{1}{v(Q(x,r))}\int_{Q(x,r)} |h|\,v\,dy \\ &\leq M^c_v f(x) + M^c_v h(x).\end{aligned}$$

The right-hand side does not depend on r, so taking the supremum over $r > 0$ yields $M^c_v(f + h)(x) \leq M^c_v f(x) + M^c_v h(x)$. For a scalar λ, the factor $|\lambda| \geq 0$ can be taken out of the supremum, implying

$$M^c_v(\lambda f)(x) = \sup_{r>0} \frac{1}{v(Q(x,r))}\int_{Q(x,r)} |\lambda|\,|f|\,v\,dy = |\lambda|\,M^c_v f(x).$$

Hence M^c_v is sublinear.

By Lemma 5, $v(\{M^c_v f > t\}) \leq \frac{N_d}{t}\|f\|_{L^1_v}$ for every $f \in L^1_v(\mathbb{R}^d)$ and $t > 0$. For $f \in L^\infty_v(\mathbb{R}^d)$, (17) with $f_2 = f$ shows that every average in the definition of $M^c_v f(x)$ is at most $\|f\|_{L^\infty_v}$. Hence $M^c_v f(x) \leq \|f\|_{L^\infty_v}$ for every $x \in \mathbb{R}^d$, and in particular $\|M^c_v f\|_{L^\infty_v} \leq \|f\|_{L^\infty_v}$.

Every $f \in L^q_v(\mathbb{R}^d)$ satisfies $f = f\chi_{\{|f|>1\}} + f\chi_{\{|f|\leq 1\}} \in L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$. The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(y)\,dy)$ with exponents 1 and ∞, now gives the claim with a constant depending only on q and on the constants N_d and 1, and hence only on q and d. □

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